DESIGN AND IMPLEMENTATION OF A MACHINE LEARNING-POWERED SURVEILLANCE SYSTEM FOR PREDICTING INFECTIOUS DISEASES: MALARIA AND CHOLERA IN KENYA

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DECLARATION

I hereby declare that this research proposal is my original work and has not been presented for a degree in any other University.

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Signature Date

This proposal has been submitted for examination with my approval as University Supervisor.

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ABSTRACT

Infectious diseases, specifically malaria and cholera, remain significant public health challenges in Kenya, with 75% of the population at risk of malaria and recurrent cholera outbreaks intensifying in underserved areas (World Health Organization, 2024). Despite the digital transformation under the Digital Health Act (2023), infectious disease management remains trapped in a reactive cycle. Current surveillance systems, such as DHIS2 and HMIS, rely on "lagging indicators"—recording cases only after patients present at clinics—meaning that by the time data is aggregated and analyzed, outbreaks are often already in an exponential growth phase. There is a critical research gap in developing an integrated, automated framework that seamlessly merges real-time climate data with routine health records to provide simultaneous early warning signals for both diseases across all 47 counties in Kenya.

This research proposes a full-stack machine learning-powered surveillance system designed to transition disease monitoring from reactive reporting to proactive prediction. The system utilizes a unified platform to synthesize historical health records, climatic data (rainfall, temperature, humidity), and population density to generate actionable "risk scores". By integrating Long Short-Term Memory (LSTM) neural networks for time-series forecasting and Geographic Information Systems (GIS) for geospatial visualization, the platform will enable health authorities to identify and visualize transmission "hotspots" in real-time.

The project adopts an Agile-Scrum system development approach, following iterative phases of requirements gathering, data preprocessing, and implementation in sprints.

The expected outcome is a web-based predictive analytics platform that provides interactive dashboards for public health officials. The system will deliver validated, county-specific forecasts, allowing for the proactive allocation of medical resources such as vaccines and antimalarials—before an outbreak reaches its peak. Ultimately, this research aims to provide a scalable, intelligent health system that reduces morbidity and mortality by enhancing the speed and accuracy of infectious disease detection in Kenya.

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Acronyms

Acronym	Definition
ADASYN	Adaptive Synthetic Sampling
ANN	Artificial Neural Network
ARIMA	Adaptive Reference Machine Learning
AUC	Area Under the Curve
CHAP	Climate Health Analytics Platform
CNN	Convolutional Neural Network
CORP	Cholera Outbreak Risk Prediction
DHIS2	District Health Information Software 2
DL	Deep Learning
DT	Decision Trees
EHR	Electronic Health Record
EPSG	European Petroleum Survey Group (coordinate system reference)
HIE	Health Information Exchange
HMIS	Health Management Information System
IDSR	Integrated Disease Surveillance and Response
KeNADA	Kenya National Data Archive
KHIS	Kenya Health Information System
K-NN	K-Nearest Neighbors
LR	Logistic Regression
MLP	Multilayer Perceptron
MSE	Mean Squared Error
NCDC	Nigerian Centre for Disease Control
NiMet	Nigerian Meteorological Agency
PCA	Principal Component Analysis
PEM	Predictive Epidemic Modeling
RFR	Random Forest Regression
RLR	Regularized Linear Regression
RNN	Recurrent Neural Network
SMOTE	Synthetic Minority Oversampling Technique

SVM	Support Vector Machine
WASH	Water, Sanitation, and Hygiene
XGBoost	Extreme Gradient Boosting

Definition of Terms

Agile-Scrum: A system development methodology that combines Agile's iterative and flexible approach with Scrum's structured sprint-based framework, used to guide the development of the proposed surveillance system.

Algorithm: A mathematical procedure used to identify patterns in data during the machine learning process.

Artificial Intelligence (AI): A branch of computer science that enables machines to simulate human intelligence, including learning, reasoning, and decision-making, used in this study for disease outbreak prediction.

Class Imbalance: A condition in epidemiological datasets where negative cases (no outbreak) significantly outnumber positive cases (outbreak), which can bias machine learning model predictions.

Climatic Drivers: Physical climatic conditions such as temperature, rainfall, and relative humidity that influence the spread and intensity of malaria and cholera outbreaks.

Compartmental Model: A mathematical model that divides a population into distinct groups based on disease status, such as Susceptible, Infected, and Recovered (SIR), to understand disease transmission dynamics.

Data Preprocessing: The process of cleaning, normalizing, formatting, and integrating raw data into a unified pipeline to prepare it for machine learning model training.

Deep Learning (DL): A specialized sub-field of machine learning that uses multi-layered artificial neural networks to solve complex, non-linear problems such as disease outbreak prediction and medical imaging.

Disease Surveillance: The continuous and systematic collection, analysis, and interpretation of health data used to monitor and respond to disease trends and outbreaks in a population.

Early Warning System: A proactive mechanism that detects and signals the likelihood of a disease outbreak before it reaches its peak, enabling timely public health intervention.

Endemic Zone: A geographic area where a disease such as malaria is constantly present or regularly found among the population.

Feature: An individual measurable input variable or predictor used by a machine learning model to make predictions, such as rainfall, temperature, or population density.

Full-Stack System: A complete software system that encompasses both the frontend user interface and the backend server, database, and machine learning components working together as a unified platform.

Geographic Information System (GIS): A computer-based system used to capture, store, analyze, and visualize spatial or geographic data, used in this study to map and display disease outbreak hotspots across Kenyan counties.

Geospatial Data: Data that contains geographic or location-based information, used in this study to map disease distributions and identify high-risk transmission areas.

Hotspot: A geographic area identified as having a significantly higher risk or concentration of disease transmission compared to surrounding areas.

Label (Target Variable): The specific outcome a machine learning model is trained to predict, in this case the risk level of a malaria or cholera outbreak classified as either high risk or low risk.

Lagging Indicator: A measure that records events only after they have already occurred, such as disease cases recorded only after patients present at health facilities, limiting early outbreak detection.

Long Short-Term Memory (LSTM): A type of recurrent neural network architecture specifically designed to learn patterns in sequential and time-series data, used in this study to forecast disease outbreak trends over time.

Machine Learning (ML): A sub-field of Artificial Intelligence that enables algorithms to learn directly from data and make predictions or decisions without being explicitly programmed for specific tasks.

Morbidity: The incidence or prevalence of a disease or medical condition within a population, used in this study as a measure of the burden of malaria and cholera in Kenya.

Mortality: The number of deaths caused by a disease within a population over a given period, used as an indicator of the severity of malaria and cholera outbreaks in Kenya.

Outbreak: A sudden increase in the number of cases of a disease above what is normally expected in a specific area or population within a given time period.

Overfitting: A condition where a machine learning model learns the specific patterns and noise of the training data too closely, resulting in poor performance when applied to new, unseen data.

Predictive Analytics: The use of statistical algorithms, machine learning models, and historical data to forecast future events or trends, in this case the likelihood of disease outbreaks.

Public Health Informatics: An interdisciplinary field that applies information science, computer science, and public health principles to improve the collection, management, and use of health data for decision-making.

Real-Time Data: Data that is collected, processed, and made available for use immediately or with minimal delay, enabling timely monitoring and response to disease outbreaks.

Risk Score: A numerical value generated by the predictive model that indicates the likelihood of a disease outbreak occurring in a specific geographic area based on environmental and health data inputs.

Supervised Machine Learning: A type of machine learning where a model is trained using labeled historical datasets containing known input variables and expected outputs, to learn patterns and make predictions on new data.

Time-Series Forecasting: A technique used to predict future values based on previously observed data points collected over time, applied in this study to forecast disease outbreak trends using historical health and climate data.

Unsupervised Machine Learning: A type of machine learning where algorithms analyze and group unlabeled data based on inherent patterns or similarities, without relying on predefined outcomes or target labels.

Vector: In the context of infectious diseases, a living organism such as the Anopheles mosquito that transmits a pathogen such as the malaria parasite from one host to another.

Web-Based Platform: A software application or system that is accessed and operated through a web browser over the internet, used in this study to deliver the predictive analytics dashboards to public health officials

CHAPTER 1
INTRODUCTION

1.1 Background of the Study

Infectious diseases are illnesses caused by pathogenic microorganisms such as bacteria, viruses, parasites or fungi and can be transmitted from one person to the other either directly or indirectly or through environmental factors and with potential for global spread and impact (World Health Organization, 2023). Despite the advancements in medical science, infectious diseases such as malaria and cholera still pose a significant burden on global health systems and landscape, majorly in low and middle-income countries. According to the World Health Organization (WHO), infectious diseases are among the leading causes of morbidity and mortality globally, with malaria and cholera being notable examples. Research has shown that the major factors contributing to the increasing prevalence of these diseases include climate change, globalization and environmental factors (Siiba et al., 2024). Globally, health organizations have adopted disease surveillance systems and digital platforms to monitor, detect, and respond to outbreaks of infectious diseases (World Health Organization, 2022). However, many of these systems and platforms remain to be largely reactive, relying on historical data and delayed reporting, which limits their effectiveness in early disease outbreak, detection and prevention.

In Kenya, infectious diseases like malaria and cholera pose significant public health challenges, heavily impacting vulnerable populations. The Ministry of Health (MoH) in Kenya plays a crucial role in managing these diseases through surveillance, prevention, containment and treatment strategies. Despite these efforts, persistent gaps in the public health systems hinder effective prediction and outbreak management of such infectious diseases. The Kenya Ministry of Health relies on the Integrated Disease Surveillance and Response (IDSR) framework, supported by the Health Management Information System (HMIS) and the District Health Information Software (DHIS2), primarily relying on health facility reporting and community-based surveillance. However, these systems often suffer from delays, underreporting, and lack of real-time data, hampering effective outbreak management.

By 2026, the focus of Kenyan public health system has shifted toward Predictive Analytics. The introduction of the DHIS2, Climate Health Analytics Platform (CHAP) and collaborative efforts between the Kenya Meteorological Department and KEMRI (Kenya Medical Research Institute) mark a transition toward using Machine Learning (ML) to forecast disease risks based on climate variables (e.g., rainfall anomalies and temperature deviations). However, there remains a critical research gap in developing an integrated, automated framework that seamlessly merges real-time climate data with routine health facility records to provide actionable early warning signals for both Malaria and Cholera simultaneously (Bukhair & Ajisafe, 2025).

1.2 Project Overview

This study is situated within the multidisciplinary field of Public Health Informatics and Data Science, specifically focusing on the development of Predictive Epidemic Modeling (PEM). It aims to bridge the gap between traditional epidemiology and modern computational intelligence by creating a full-stack analytics framework tailored for infectious disease surveillance.

Globally, the landscape of infectious disease surveillance is undergoing a paradigm shift. In 2026, the artificial intelligence (AI) market for disease outbreak prediction is estimated to exceed $3 billion, driven by the increasing frequency of climate-induced health crises. International health bodies, including the WHO Hub for Pandemic and Epidemic Intelligence, are advocating for the integration of "Big Data"—encompassing environmental sensors, mobility patterns, and real-time electronic health records (EHRs) to move beyond reactive reporting. Advanced nations have begun implementing cloud-native, AI-driven platforms like BlueDot and HealthMap, which scan global data to identify emerging threats weeks before traditional clinical detection.

In Kenya, the post-COVID-19 era and the enactment of the Digital Health Act (2023) have laid the foundation for a national Health Information Exchange (HIE). Despite this, regional health surveillance remains primarily descriptive. For instance, while the Kenya Meteorological Department and KEMRI have launched early warning systems for the Western Highlands, these models are often siloed and lack a unified "full-stack" interface for county-level health officers. Malaria and Cholera continue to be exacerbated by erratic rainfall patterns and urban density, highlighting a critical need for a localized, integrated platform that translates complex data into actionable public health "risk scores" across all 47 counties.

The proposed system aims at utilizing a full-stack architecture to integrate data from various sources such as historical health data, climatic and environmental factors. The proposed system will deploy Machine Learning & Time-Series Forecasting models such as Long-Short Term Memory to identify patterns between environmental anomalies (temperature/rainfall) and disease spikes. It aims to integrate Geospatial metadata that is Geographical Information Systems coordinate system (EPSG:4326) for mapping county boundaries and provide visualization of real-time risk maps. The proposed system will enable a proactive intervention strategy, ensuring that medical resources are deployed before an outbreak reaches its peak.

1.3 Statement of the problem

Infectious diseases continue to pose a significant global public health challenge despite advancements in medical research and surveillance systems. According to the World Health Organization, there were approximately 282 million cases of Malaria and about 610,000 related deaths worldwide in 2024, with over 90% of the burden (265 million) occurring in Africa (WHO, 2025). Similarly, Cholera remains a major global public health concern, with more than 560,000 cases and over 6000 deaths reported in 2024, particularly in regions with inadequate water and sanitation systems (WHO, 2024). These statistics demonstrate the persistent and widespread impact of infectious diseases globally.

In Kenya, the burden of infectious diseases remains high, with 75% of the population at risk of the disease. Malaria is endemic in many parts of the country, with an estimated 3.3 million cases and over 1,000 deaths reported annually (WHO, 2024). Cholera outbreaks also occur periodically, especially in densely populated and underserved areas, contributing to increased morbidity and strain on healthcare systems. The country reported approximately 8,926 cases and 145 deaths in 2023, with outbreaks intensifying in 2022-2023, exceeding 12,000 cases and over 200 deaths (Kiama et al., 2025). Despite efforts by the Ministry of Health and county governments to strengthen disease surveillance, the existing systems are largely reactive, relying on historical data and delayed reporting mechanisms.

Despite the digital transformation of Kenya's healthcare sector under the Digital Health Act (2023), infectious disease management remains trapped in a reactive cycle. Additionally, the reliance on reactive surveillance systems such as the DHIS2 and HMIS results in delays in outbreak detection, limited real-time insights, and ineffective early warning capabilities. Studies have shown that surveillance systems in many developing countries suffer from incomplete and untimely reporting, which significantly hinders timely response to disease outbreaks (Malebana et al., 2025). Consequently, outbreaks are often identified after they have already spread, leading to increased transmission rates, higher healthcare costs, and preventable loss of lives.

The magnitude of this problem is evident in the frequent recurrence of outbreaks, overstretched healthcare facilities, and reduced effectiveness of intervention strategies. Currently, surveillance systems rely on "lagging indicators"—cases are only recorded after patients present at health facilities. By the time this data is aggregated and analyzed at the county level, an outbreak is often already in its exponential growth phase. Furthermore, current surveillance approaches do not adequately leverage modern technologies such as machine learning and data analytics to predict outbreaks or identify high-risk areas. This limits the ability of health authorities to make proactive, data-driven decisions toward public health response.

Therefore, the core research problem lies in the absence of an adequately integrated, full-stack machine-learning predictive analytics platform that leverages Long Short-Term Memory (LSTM) networks and Geospatial metadata (EPSG:4326) to synthesize historical health records with real-time environmental variables, capable of predicting and visualizing malaria and cholera outbreaks in Kenya. Addressing this gap is essential to enhance early detection, improve response time, and reduce the overall impact of infectious diseases, particularly malaria and cholera in Kenya.

1.4 Proposed Solution

To address this challenge, this study proposes a machine learning-powered surveillance system designed to predict and visualize infectious disease outbreaks, specifically for malaria and cholera across Kenyan counties, leveraging Long Short-Term Memory (LSTM) neural networks and geospatial mapping. This system will integrate multiple data sources, including climatic data, population density, and historical outbreak records, to develop predictive models to forecast potential outbreaks, providing real-time risk assessments to aid public health decision-making. The proposed system will also provide interactive dashboards for health officials to visualize outbreaks and allocate resources efficiently, ultimately improving public health outcomes.

1.5 Objectives

1.5.1 General Objective

The main aim of this research proposal is to develop a full-stack machine learning-powered surveillance system for predicting and visualizing the outbreak of malaria and cholera across Kenyan counties.

1.5.2 Specific Objectives

1. To collect and preprocess multi-source data, including historical health data, environmental, and geospatial data, to identify key factors influencing outbreaks of Malaria and Cholera in Kenya.
2. To design a full-stack system architecture and a machine learning-based predictive model for forecasting the risk of malaria and cholera outbreaks.
3. To develop a web-based predictive analytics system for visualizing disease outbreak risks at the county level in Kenya.
4. To evaluate the performance of the predictive model using historical datasets and appropriate metrics such as prediction accuracy and model validation techniques.

1.6 Research Questions

1. How does historical health data, environmental, and geospatial data influence the outbreak of Malaria and Cholera in Kenya?
2. How can a full-stack system architecture and machine learning models be optimized to process multi-dimensional geospatial and health data for reliable outbreak forecasting?
3. In what ways can a web-based visualization framework improve the interpretability of complex predictive analytics for public health decision-makers at the county level?
4. What is the predictive reliability and accuracy of the developed machine learning models when validated against historical outbreak data using standard statistical error metrics?

1.7 Justification

Infectious diseases continue to pose a significant burden to global health systems and within Kenya as well. This underscores the urgent need for the adoption of more effective predictive and proactive infectious diseases surveillance systems. Infectious diseases such as Malaria and Cholera still record significant morbidity and mortality rates, especially in low and medium-income countries with limited healthcare infrastructure and poor sanitation. This is despite the existence of surveillance systems which rely on "lagging indicators" (reporting cases only after they appear in clinics), retrospective data and delayed reporting mechanisms, limiting their ability to effectively provide predictive early warnings to support timely response and interventions.

This research is justified by the need to address the limitations posed by the existing Malaria and Cholera surveillance systems which are insufficient in the prediction and visualization of the infectious diseases. By leveraging emerging technologies such as machine learning and data analytics, the proposed system aims to transform disease surveillance from a reactive to a predictive approach. This will enable early detection of potential outbreaks, allowing health authorities to implement preventive measures before widespread transmission occurs.

The proposed solution will directly benefit key stakeholders, including the Ministry of Health, county health departments, vulnerable populations, and public health practitioners, by providing a tool that enhances decision-making through real-time risk prediction and visualization. Additionally, communities will benefit from improved outbreak preparedness and reduced exposure to infectious diseases. The system also contributes to more efficient allocation of healthcare resources by identifying high-risk areas in advance.

From a research perspective, this study contributes to the growing field of intelligent health systems by integrating machine learning techniques with spatial data analysis for disease prediction. While previous studies have explored disease surveillance, there is limited implementation of integrated predictive systems within the Kenyan context. Therefore, this research fills an important gap by demonstrating how advanced technologies can be applied to improve public health outcomes in developing countries.

Furthermore, the relevance of this study is underscored by current global trends emphasizing the use of digital technologies and artificial intelligence in healthcare. As the world continues to face emerging and re-emerging infectious diseases, there is a critical need for innovative, data-driven and predictive approaches to disease monitoring and control. This study is therefore timely and aligns with ongoing efforts to strengthen health systems through technological innovation.

1.8 Proposed System Methodologies

The proposed methodology for the development of this system will follow the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which is widely used in data science and machine learning projects. This methodology is suitable for this project because it provides a structured, iterative approach specifically tailored for data-driven systems such as the prediction and visualization of infectious disease outbreaks. CRISP-DM emphasizes continuous refinement, understanding of data, and alignment between technical processes and real-world objectives, making it ideal for developing a predictive analytics platform for Malaria and Cholera outbreaks in Kenya. The development will take place in the following phases iteratively:

1.8.1 Business Understanding

This phase will involve clearly defining the objectives of the system in relation to public health needs. The goal of the project is to develop a machine learning-powered surveillance system capable of predicting and visualizing Malaria and Cholera outbreaks across Kenya. During this phase, key stakeholders such as public health officials will be identified, and their requirements will be gathered to ensure the system addresses real-world challenges. The expected outcomes, including early outbreak detection, improved decision-making, and effective resource allocation, will also be established.

1.8.2 Data Understanding

In this phase, relevant data sources will be identified and collected to support the prediction of disease outbreaks. This will include historical disease records, weather data, population density statistics, and environmental factors that influence the spread of Malaria and Cholera. The collected data will be explored to understand its structure, quality, and relevance to the project objectives. Initial data analysis will be performed to identify patterns, trends, and potential issues such as missing values or imbalanced datasets.

1.8.3 Data Preparation

This phase will involve transforming the raw data into a clean and usable format suitable for machine learning. Data preprocessing tasks such as cleaning, normalization, integration, and formatting will be carried out to ensure consistency and accuracy. Tools such as Python, Pandas, and NumPy will be used to manipulate and process the datasets efficiently. Feature engineering techniques will also be applied to extract meaningful variables that improve the predictive performance of the model.

1.8.4 Modeling

During the modeling phase, machine learning techniques will be applied to develop predictive models for disease outbreaks. The project will utilize LSTM (Long Short-Term Memory) neural networks implemented using TensorFlow or Keras to handle time-series forecasting. Different model configurations and parameters will be tested and optimized to achieve the best performance. The model will be trained using the prepared dataset and evaluated iteratively to improve accuracy.

1.8.5 Evaluation

In this phase, the developed model will be thoroughly evaluated to ensure it meets the project objectives and performs effectively in predicting disease outbreaks. Performance metrics such as accuracy, precision, recall, and F1-score will be used to assess the model. In addition to technical evaluation, the system will also undergo user validation through feedback from public health officials to ensure usability and relevance. Any necessary improvements will be identified before deployment.

1.8.6 Deployment

The will be the final phase which will involve deploying the system as a full-stack web-based platform accessible to users. The system will be hosted on cloud platforms such as Heroku or Render to ensure scalability and availability. The deployment will include the integration of the machine learning model, backend APIs, database, and frontend dashboard with geospatial visualization tools like Leaflet or Mapbox. Continuous monitoring and maintenance will be carried out to update datasets, improve model performance, and ensure the system remains effective over time. Version control and collaboration will be managed using Git and GitHub to support future enhancements.

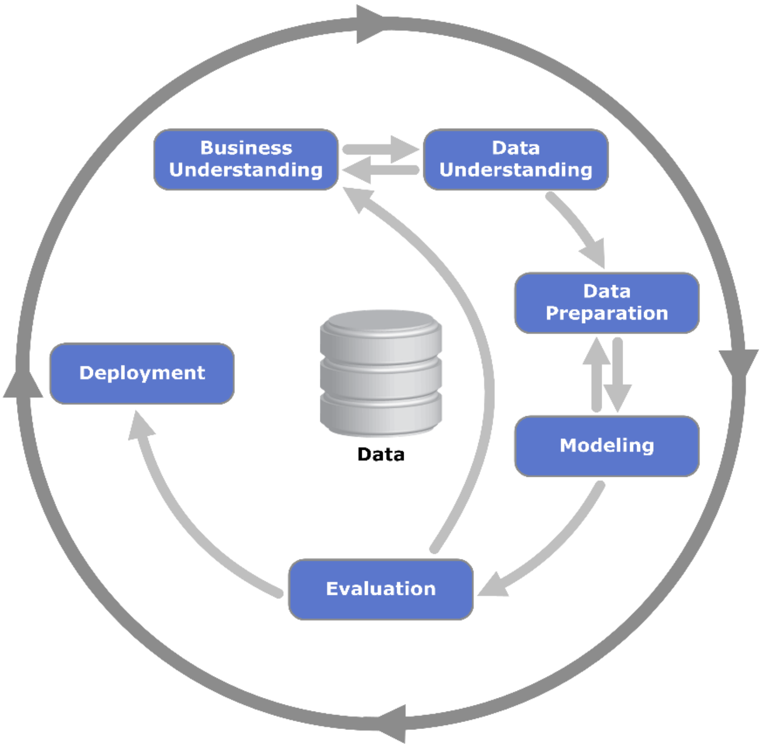

Figure 1.1 : CRISP-DM Methodology

1.9 Scope

This study focuses on the development of a machine learning-powered surveillance system to predict the outbreak of infectious diseases, specifically malaria and cholera across the Kenyan Counties. The research will make use of the available historical health datasets including, and real-time data related to climatic factors such as rainfall, temperature and humidity as well as environmental factors such as water sources and sanitation conditions.

The target population for this study are the communities living in the high-risk areas for malaria and cholera prevalence as well as public health stakeholders whom the surveillance system will help gain predictive insights and visualization of disease trends thereby helping in proactive response to the outbreaks.

The project will be limited to the development of a machine learning-surveillance system with integrated Geographical Information System visualization component for mapping the distributions of disease outbreaks and identifying potential hotspots. The study will not encompass the diagnosis of diseases, disease treatment or the pharmaceutical interventions for malaria and cholera.

Furthermore, this study acknowledges the following potential limitations in relation to the field of study:

1. Resource Constraints: This study acknowledges computational resource constraints and the constraints in the optimization of the machine learning model.
2. Environmental Determinism: the models that will be used will primarily focus on climatic and environmental drivers. As a result, socio-economic factors occurrences like the sudden changes in migrations or local sanitation conditions among communities will not be captured I the predictive model.
3. Methodological Constraint: the machine learning model will identify the relations between climatic and environmental factors and the spread of malaria and cholera, but will not prove the biological causes of these diseases.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

This chapter aims to present a comprehensive understanding and review of the existing knowledge and research related to the adoption and application of machine learning, data-driven systems and geospatial technologies in the prediction and visualization of malaria and cholera outbreaks. The review evaluates how emerging technologies such as artificial intelligence, Geographical Information Systems and predictive analytics are being adopted in disease surveillance and decision-making to enhance public health response in the management of the outbreak of these diseases.

This study focuses on the adoption and application of machine learning techniques and models in the prediction of infectious diseases, specifically malaria and cholera in Kenya. These diseases have been monitored and their outbreaks managed largely using traditional surveillance systems which have posed challenges such as being slow, reactive and significant reporting lags. The integration and application of machine learning techniques and models addresses the systematic vulnerabilities of traditional surveillance systems through its capability to analyze vast, heterogenous datasets including satellite-based climate data socio-economic indicators, and electronic health records—to identify complex patterns and anomalies indicative of an outbreak (Tshimula et al., 2024). Machine learning and Artificial Intelligence integration into the surveillance of infectious diseases highlights a remarkable shift in the prediction and management of the outbreaks of malaria and cholera particularly in regions with limited resources like the sub-Saharan Africa (Tshimula et al., 2024).

There has been a growing adoption and application of machine learning models, geospatial and real-time predictive analytics to improve the forecasting and visualization of the outbreaks of malaria and cholera in Kenya. Machine learning (ML) and Artificial Intelligence (AI) are particularly valuable in this setting and are becoming rapidly applied in healthcare since these technologies can analyze vast data collections to uncover insights that are hidden which are missing in traditional statistical methods (Muriithi et al., 2024). Machine Learning models provide the ability to draw knowledge from data and identify relevant patterns using classification methods. The application of Machine Learning algorithms can be of value to healthcare providers in the identification of effective measures and in the decision-making process for the best public health response (Nkiruka et al., 2021).

This literature review aims to evaluate the existing studies and systems that are related to the prediction and visualization of the outbreak of infectious diseases such as malaria and cholera. The review also serves a purpose in identifying and assessing the methodologies and technologies that have been used in the application of machine learning to predict these outbreaks, as well as identifying the gaps that the proposed system seeks to address. Particularly, this literature review seeks to:

1. Examine the existing surveillance systems and approaches to the prediction of malaria and cholera outbreak.
2. Asses the application of machine learning, artificial intelligence and GIS visualization in disease prediction and surveillance.
3. Identify the limitations and shortcomings of the existing prediction and surveillance systems.
4. Justify the need for a more integrated, rapid data-driven framework, real-time, and geospatially enabled predictive platform for the prediction of malaria and cholera outbreak in Kenya.

2.2 Theoretical Review

This section presents the core concepts forming the foundation of Machine Learning applied in the development of a prediction and visualization system of malaria and cholera. It also assesses the key theoretical divisions defining the application of machine learning in the prediction of infectious diseases. Additionally, it highlights the advantages and shortcomings of each kind of the theoretical divisions defining this field, particularly in the prediction and analytics of the outbreaks of infectious diseases.

2.2.1 Key Concepts and Variables

Several core and foundational machine learning concepts are defined as follows to provide an understanding of the design and development of a machine learning-powered surveillance system for malaria and cholera. The key environmental and socio-economic variables that are critical in the prediction of the outbreaks of malaria and cholera are as well defined.

1. **Machine Learning (ML)**

   Machine Learning is the sub-field of Artificial Intelligence that deals with the construction of algorithms that are capable of learning directly from data without relying on other predetermined equations (Leo, 2020). An algorithm is the mathematical procedure used to identify patterns in data.

1. **Data**

   Data refers to raw facts and figures collected from various sources such as historical health records, climatic data, and population statistics. In this study, data includes malaria and cholera case records, rainfall levels, temperature, humidity, population statistics, and sanitation indicators.

2. **Features (Predictors)**

   Features are the input parameters or individual independent variables used by a model to make a prediction (Sadiq et al., 2024). In the Kenyan context, key features include climatic factors such as rainfall, temperature, humidity, geographic data such as endemic zones, altitude, and socio-economic indicators such as access to basic sanitation, population density.

3. **Labels (Target Variable)**

   The label is the specific outcome the model is designed to predict (Muriithi et al., 2024). The label in this case is the risk level of disease outbreak which is either high risk or low risk for malaria and cholera outbreak.

4. **Training and Testing**

   Training involves feeding historical data into a model so its algorithm can learn patterns, while testing evaluates how well the model performs on new, unseen data.

5. **Performance Metrics**

   Performance metrics the are quantitative measures of model performance that are used to measure the effectiveness of a surveillance system. The metrics that will be used include:

   i. Accuracy: Refers to the overall proportion of correct predictions that are made by the machine learning model (Muriithi et al., 2024).
   ii. Sensitivity: This refers to the ability of the model to identify true positive cases correctly, which is important in the medical/healthcare context to avoid missing actual infections (Leo, 2020).
   iii. Specificity: This is the ability of the prediction model to identify individuals who are disease-fee that is the true negatives (Muriithi et al., 2024).

iv. F1-Score: The weighted harmonic mean of precision and recall, providing a balanced measure for models tested on imbalanced datasets (where one class significantly outnumbers the other) (Muriithi et al., 2024)

6. **Geographic Information Systems (GIS)**

   GIS refers to computer systems used to capture, store, analyze, and visualize spatial or geographic data. In this study, GIS is used to map and visualize disease outbreaks across different counties in Kenya.

7. **Climatic Drivers**

   These are the physical climatic conditions that form primary determinants in influencing the spread of malaria and cholera in the context of the prediction of these diseases. They include temperature, precipitation (rainfall), and relative humidity.

8. **Weather Drivers**

   These are the climatic factors that influence the occurrence, spread, and intensity of events or conditions especially in the outbreaks of diseases. For example, higher temperatures and excessive rainfall favor the growth of *Vibrio cholerae* which is the is the bacterium responsible for the infectious disease cholera.

2.2.2 Theoretical Frameworks for Infectious Disease Prediction

2.2.2.1 Statistical and Epidemiological Models

Statistical/Epidemiological Models are defined as the logical and mathematical demonstrations of diseases epidemiology and their related processes (Çağırgan & Çağırgan, 2020). These models also rely on statistical relationships to describe disease dynamics and identify key factors influencing outbreaks of diseases such as malaria and cholera. The models focus on identifying and ranking the mechanisms that are responsible for disease transmission and provide the necessary tools for the interpretation of studies related to epidemiology (Jewell, 2003). This school of thought relies on formalising the relationships between variables through fixed mathematical equations (Inam, 2025). These models are widely used in epidemiology for analysis, forecasting, and public health planningand decision-making. While traditional models such as mathematical, statistical and epidemiological models have heavily relied on differential equations and compartmental structures, recent studies highlight a paradigm shift toward incorporating advanced machine learning (ML) techniques to enhance the predictive accuracy (Inam, 2025).

The Statistical/Epidemiological Models are further classified into several broad categories based on their objectives and structures as follows:

i. **Compartmental Models:** Compartmental models are a subset of mathematical models that divide/partition a population into distinct groups known as compartments based on disease status. A common framework used in the compartmental models is the SIR framework which is expressed as:

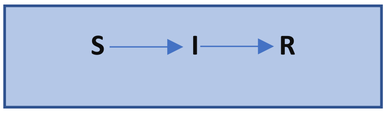

Figure 2.1: SIR Framework

Where:

S (Susceptible): Individuals who can contract the disease.

I (Infected): Individuals who have the disease and can transmit it.

R (Recovered): Individuals who have recovered and may have immunity.

Disease transmission occurs through interactions between compartments, and individuals transition from one state to another based on defined rates. Compartmental models are used to understand the spread of diseases like cholera (through contaminated water) and malaria (through vector transmission), helping to estimate outbreak magnitude and duration.

ii. **Deterministic and Stochastic Models:** Deterministic models assume that disease predictions follow fixed mathematical relations between different elements and provide the same result for the same starting conditions meaning that the same input will always produce the same output (Çağırgan & Çağırgan, 2020). Stochastic models include elements that are random and have chances, which makes them more realistic since they reflect the variability of nature, also providing a range of possible outcomes (Çağırgan & Çağırgan, 2020).

iii. **Static and Dynamic Models:** Static models represent the condition of a system at a given point in time, while dynamic models account for changes that have taken place over a specific interval. Dynamic modelling is preferred for scenarios like vaccination, where incidence varies over time before settling (Çağırgan & Çağırgan, 2020).

iv. **Network Models:** These model view individuals as nodes and their interactions as edges in a graph. Network models represent the spread of diseases as a process that occurs through connections or relationships (edges) between individuals or locations **(**nodes**)**. Instead of assuming everyone mixes equally, this approach models who interacts with whom, making it more realistic for transmission dynamics (Craig et al., 2020).

Advantages of Statistical and Epidemiological Models

i. **Predictive Power and Decision Support:** the models enable public health officials to assess the behavior beforehand using presumptive scenarios, therefore assisting in the development of vaccination, treatment, and public health strategies (Çağırgan & Çağırgan, 2020).
ii. **Uncertainty Elimination:** The models provide ways of overcoming uncertainties by combining experimental data, reviews by experts on the dynamics of infections and field of knowledge to predict uncertainties in the occurrence of infectious diseases.
iii. **Interventions Evaluation:** These models are crucial in the testing of various options of control (e.g., patient isolation, face mask policies, the closure of schools) to identify which measure are most effective at decreasing the spread of infectious diseases.
iv. **Analytical Clarity:** The models enable the microscopic explanation of macroscopic disease behaviour and help measure the uncertainties inherent in future predictions (Çağırgan & Çağırgan, 2020).

Limitations of Statistical and Epidemiological Models

i. **Data Quality Issues:** These models are heavily constrained by data quality and availability. Issues like underreporting, misdiagnosis, and reporting delays (where data trickles in over several days) can lead to spurious conclusions.
ii. **Usability Barriers:** Traditional mathematical models are often complex to understand and extend for non-mathematicians, requiring deep knowledge of model equations.

iii. **Assumption Sensitivity:** Simple models (like the basic SIR) often make unrealistic assumptions, such as homogeneous mixing or immediately starting infectiousness, which can dwarf noise-related uncertainty.

iv. **Dependent Happening:** Unlike non-communicable diseases (e.g., lung cancer), observation of one infected individual is not independent of others in communicable diseases, which confuses the assessment of population effectiveness.

2.2.2.2 Supervised Machine Learning

Recent studies have identified Supervised Machine Learning as being the most efficient prediction paradigm for predicting climate-sensitive diseases as well as infectious diseases, particularly in resource-limited countries like Sub-Saharan Africa (Inam, 2025). Supervised machine learning is defined by the use of labeled datasets containing historical input data such as climatic or socio-economic variables and target labels (such as "positive" or "negative" test results) to guide the algorithm's learning process (Ileperuma et al., 2023). It involves training a model using labeled historical data, where the input variables (features) and the expected output (label) are already known. The model is trained using historical datasets and then tested to evaluate its prediction accuracy. By analyzing past data, the model identifies patterns and relationships that can be used to make predictions on new, unseen data (Hu et al., 2025). In disease prediction, this means using historical data (e.g., climate, demographics, past cases) to predict whether or when diseases like malaria or cholera will occur.

Recent studies demonstrate that supervised ML consistently outperforms traditional statistical and mathematical models in providing quick and accurate disease forecasts (Tshimula et al., 2024). The following recent studies have explained the performance of Supervised Machine learning in the prediction of diseases:

1. **Malaria Prediction:** A 2024 study in Kenya found that the Random Forest algorithm was the most robust for predicting malaria occurrence, achieving a classification accuracy of 97.33% (Muriithi et al., 2024). In Senegal, Random Forest models reached accuracy levels above 0.9 in regions with high prevalence by analyzing satellite-derived rainfall data (Ileperuma et al., 2023). Additionally, ML has been used to predict mosquito bed net utilization with 83% accuracy (Tariq et al., 2025).
2. **Cholera Outbreak Forecasting:** The Cholera Outbreak Risk Prediction (CORP) model, utilizing supervised techniques, achieved a remarkable 99.62% accuracy in

Nigeria (Tshimula et al., 2024). Other 2024 research in Nigeria using Random Forest achieved 92% accuracy and integrated user-friendly interfaces for real-time alerts (Omankwu & Etuk, 2024).

3. **COVID-19 Tracking:** In Zambia, the XGBoost algorithm was identified as the best performer for predicting mortality in hospitalised patients, reaching 92.3% accuracy (Tshimula et al., 2024). In South Africa, Artificial Neural Networks (ANN) and Random Forest were used to identify advanced age and severe symptoms as top predictors of ICU mortality (Tshimula et al., 2024).

Advantages of Supervised Machine Learning

i. **Accuracy and Robustness:** Models learn directly from historical labeled data, which improves reliability. Tree-based ensemble models, such as Random Forest and XGBoost, are noted for being comparatively insensitive to covariation among predictors and for delivering superior results on performance measures like AUC-ROC (Inam, 2025).

ii. **Early Warning and Proactive Response:** Can predict outbreaks before they happen using past trends. ML models facilitate real-time analytics, allowing health authorities to allocate resources (such as vaccines and antimalarials) to "hotspots" before an outbreak peaks (Inam, 2025).

iii. **Data Integration:** These algorithms can process vast, heterogeneous datasets, including satellite imagery, genomic data, and electronic health records, to uncover hidden patterns that miss in traditional methods (Inam, 2025).

iv. **Handling Complexity:** Unlike traditional models (e.g., ARIMA or SIR), supervised ML can capture complex, non-linear interactions between climate variables and disease incidence (Inam, 2025).

Limitations of Supervised Machine Learning

Despite its potential, several barriers hinder the widespread adoption of supervised learning in low-resource settings:

i. **The "Black Box" Problem:** High-performing models, especially deep learning and complex ensembles, often lack interpretability. This lack of transparency can erode trust among policymakers who must make critical decisions based on model outputs (Obeagu & Ogenyi, 2025).

ii. **Data Quality and Dependency:** These systems rely on large volumes of high-quality labeled data (Inam, 2025). In many African contexts, health information systems are fragmented, and underreporting can lead to biased model conclusions (Obeagu & Ogenyi, 2025).

iii. **Class Imbalance:** Most epidemiological datasets contain a vast majority of negative cases (e.g., 96% negative). Without advanced sampling techniques like SMOTE or ADASYN, supervised models tend to be biased toward the majority class, failing to predict actual outbreaks (Leo, 2020).

iv. **Overfitting:** There is a constant risk that a model will learn the specific "noise" of the training data rather than generalisable patterns, leading to poor performance on new datasets (Sadiq et al., 2024).

2.2.2.3 Unsupervised Learning

The main principle of unsupervised learning is that the algorithm analyzes input data and automatically groups or organizes it based on similarities or patterns without prior knowledge of outcomes. Since 2022, the application of unsupervised machine learning (ML) in disease surveillance has been recognised as a vital tool for exploratory data analysis, particularly when dealing with the high-dimensional and often unlabelled datasets common in low-resource settings (Ileperuma et al., 2023). Unsupervised learning differs from supervised methods by seeking out hidden patterns, structures, or groupings within data without relying on target labels. In disease outbreak prediction, recent studies have shown that its role expands beyond data exploration:

1. **Hotspot Identification and Clustering:** Hotspot Identification and Clustering: Algorithms like K-means are increasingly used to cluster regions by infection rates, providing insights that support effective resource allocation and targeted interventions (Inam, 2025). For instance, in Kenya, recent efforts have used these techniques to map cholera hotspots by integrating epidemiological persistence with Water, Sanitation, and Hygiene (WASH) indicators (Wamae et al., 2026).
2. **Dimensionality Reduction:** Techniques such as Principal Component Analysis (PCA) are employed to handle "excessive features" in complex datasets, identifying

the most critical factors influencing disease incidence—such as climatic and socio-economic variables—before supervised models are applied (Inam, 2025).

3. **Medical Imaging:** Recent research highlights the use of unsupervised feature learning for cystoscopic image classification and the unsupervised segmentation of 3D medical images based on clustering and deep representation learning (Intorne et al., 2023).
4. **Diarrheal Disease Determinants:** A 2023 review noted the use of **cluster analysis** to determine the socio-economic and environmental drivers of diarrheal diseases in resource-poor settings.
5. Vector Surveillance: In Iran (2025), a two-stage monitoring approach used Random Forest and MLP (often informed by unsupervised feature extraction) to identify high-risk depressions for mosquito breeding habitats, achieving an AUC score of 94%.

Advantages of Unsupervised Machine Learning

i. Pattern Discovery: Capable of discovering patterns and groupings in datasets that lack labels, which is essential for novel or emerging zoonoses.
ii. Exploratory Power: Vital for identifying meaningful trends that inform the derivation of new research questions and future analyses.
iii. Resource Strategy: Effective at clustering geographical areas by risk level, allowing health officials to station staff proactively during climate emergencies.

Limitations of Unsupervised Machine Learning

i. Interpretability: These models often lack interpretability, making it difficult for public health decision-makers to understand the rationale behind a specific cluster
ii. Validation: It is significantly harder to validate results compared to supervised learning because there is no "ground truth" or historical label to compare against.
iii. Accuracy: For direct outbreak prediction (Yes/No), unsupervised methods generally underperform compared to supervised models.

2.2.2.4 Deep Learning (Architectural Intelligence)

Deep learning (DL) is a specialized sub-field of Artificial Intelligence (AI) and Machine Learning (ML) that utilizes multi-layered artificial neural networks to solve complex, non-linear problems (Sadiq et al., 2024). It is inspired by biological neural structures in the human brain, where layers of interconnected "neurons" process information in stages (Stewart, 2022). In this framework, each layer of the network learns to extract higher-order features from the previous layer, allowing the system to automatically recognize patterns without being explicitly programmed for specific features (Ljubic et al., 2022).

Deep learning has outperformed traditional machine learning in several domains due to its ability to handle multimodal and high-dimensional data (Stewart, 2022). It employs complex architectures like:

i. Convolutional Neural Networks (CNNs): These are specifically designed for spatial analysis and are the gold standard for medical imaging. For example, CNN architectures like MobileNetV2 and ResNet50 have achieved up to 97.06% accuracy in detecting malaria parasites from microscopic blood smear images.
ii. Recurrent Neural Networks (RNNs): These excel at capturing temporal sequences and time-dependent data. Units like Long Short-Term Memory (LSTM) are frequently used for forecasting disease outbreaks (such as COVID-19 or influenza) by analyzing how contagion unfolds over time.
iii. Transformers: Emerging architectures that excel at integrating diverse datasets (e.g., social, demographic, and environmental) to provide a complete view of disease spread.

Advantages of Deep Learning

i. Higher Accuracy: DL models can capture complex, non-linear relationships that traditional models often oversimplify.
ii. Feature Learning: They reduce the need for manual feature extraction by learning feature hierarchies directly from the data.

Limitations of Deep Learning

i. "Black Box" Nature: A significant limitation is the lack of interpretability, making it difficult for clinicians to understand the rationale behind a prediction.
ii. Resource Intensity: These models involve high computational costs and require specialized hardware (like GPUs) and massive datasets for effective training.
iii. Data Requirements: DL typically requires large-scale labeled datasets; for smaller or sparse datasets, traditional ML methods remain more feasible.

2.3 Case Study Review

This section examines the adoption of existing applications of machine learning, predictive analytics and geospatial systems in the surveillance of infectious diseases, specifically focusing on malaria and cholera outbreaks prediction and visualization. The review of the case studies highlights some of the real-world implementations, successes that were achieved and their limitations in relation to the proposed machine learning surveillance system for malaria and cholera prediction and visualization in Kenya. The following case studies have been reviewed in relation to the application of machine learning in the prediction and visualization of malaria and cholera outbreaks in Kenya.

1. **Machine Learning for Malaria Prediction in Kenya**

   This case study, based on research by Muriithi et al. (2024), evaluates the implementation of supervised machine learning to predict malaria occurrence—specifically final test results—within the Kenyan context. The study is significant because it aims to complement existing public health efforts by providing a robust, data-driven approach to mitigate malaria-related threats, which remain a major concern in Kenya's tropical and endemic zones.

   The researchers utilized a well-documented dataset from the Kenya National Data Archive (KeNADA) and investigated twelve predictor variables. The research and implementation involved estimating and comparing the performance of the following five distinct supervised machine learning algorithms:

   1. Support Vector Machines (SVM)
   2. K-Nearest Neighbors (K-NN)
   3. Random Forest
   4. Tree Bagging
   5. Boosting

   The study adopted a binary classification approach (0=Negative, 1=Positive) using repeated cross-validation (five times) to ensure the models were validated against varied data subsets. The core discussion of the results is centered on the ability of these models to analyze both linear and non-linear relationships between features, which allows for deeper insights into complex epidemiological patterns than traditional statistical methods.

Key Successes

The primary success of the study and implementation was the identification of Random Forest as the superior and most robust model for malaria classification. Key performance milestones included:

i. High Accuracy: The model attained a classification accuracy of 97.33%.
ii. Robust Metrics: The model demonstrated a specificity of 98.4% and an Area Under the Curve (AUC) of 98.3%, indicating a high ability to distinguish between infected and non-infected individuals.
iii. Critical Feature Identification: The model successfully ranked the importance of predictors, revealing that the presence of Plasmodium falciparum was the most critical feature (100% relative importance), followed by geographic region, endemic zone, and anaemic levels.
iv. Resource Optimization: By identifying high-risk regions and biological predictors, the system enables health authorities to proactively allocate resources to areas with the highest susceptibility.

Possible Misses and Limitations

While the study and implementation achieved high accuracy, it highlighted several "misses" or complexities that reflect the challenges of machine learning in healthcare:

i. Unexpected Feature Irrelevance: A notable finding was that having a mosquito bed net was found to have 0% relative importance in predicting final test results within this specific model, a result that might seem counterintuitive to traditional prevention strategies.
ii. Model Sensitivity Discrepancy: While accuracy was high, the sensitivity (recall) was lower at 71.1%, suggesting the model was slightly less effective at correctly identifying all positive cases compared to its high specificity for negative cases.
iii. Inconsistency with Previous Literature: The findings contradicted some earlier studies; for instance, while this research favored Random Forest, previous literature had found that Naïve Bayes outperformed other models in predicting outbreaks.

iv. Dataset Constraints: The study's conclusions are tied to the specific characteristics of the KeNADA dataset, and the performance of such models must be continuously monitored as new data is incorporated to maintain predictive relevance over time.

2. **Machine Learning for Cholera Outbreak Prediction in Nigeria**

This case study, based on the research by Omankwu and Enefiok (2024), evaluates the development and implementation of a data-driven framework for the early detection and prediction of cholera outbreaks in Nigeria. The study is particularly relevant given Nigeria's diverse climatic landscape—ranging from arid northern regions to humid tropical southern areas—which complicates the transmission dynamics of cholera.

The researchers developed a predictive model by integrating diverse, multimodal datasets to capture the complex drivers of cholera with the following aspects of implementation:

i. Data Integration: The system pulled historical outbreak records from the Nigerian Centre for Disease Control (NCDC), weather data (daily rainfall and temperature) from the Nigerian Meteorological Agency (NiMet), and socio-economic indicators like population density and sanitation infrastructure from the National Bureau of Statistics and local agencies.

ii. Methodology: The implementation followed a systematic pipeline of data preprocessing (handling missing values via mean substitution and regression imputation), feature engineering (creating interaction terms between rainfall and temperature), and model training.

iii. Algorithm Comparison: The study experimented with several supervised learning algorithms, including Support Vector Machines (SVM), Neural Networks, and Random Forest.

The core discussion highlights that while traditional statistical models often lack the precision required for timely interventions, Machine Learning algorithms are uniquely suited for this task due to their ability to identify non-linear patterns across climatic and demographic variables.

Key Successes

The primary success of the implementation was the superior performance of the Random Forest algorithm, which proved to be the most robust for the Nigerian context with the following:

i. High Performance Metrics: The Random Forest model achieved an accuracy of 92%, a precision of 90%, a recall of 88%, and an F1-score of 89%.
ii. Identification of Key Predictors: The model successfully identified rainfall, temperature, and population density as the most influential features for predicting outbreaks.
iii. Practical Deployment (User Interface): A significant milestone was the creation of a user-friendly interface for public health officials. This platform provides real-time alerts and geospatial visualizations through GIS technology, allowing authorities to monitor hotspots and allocate resources like medical supplies and vaccines more effectively.
iv. Validation: The model's predictions showed a strong correlation with actual historical outbreak data, validating its accuracy and reliability.

Possible Misses and Limitations

Despite the technical successes realized in this study, the study identified several critical limitations and challenges that impact the system's reliability:

i. Data Quality Dependency: The model's predictive accuracy is highly contingent on the quality and completeness of the input data. Issues with missing information and data inconsistencies in regional records remain a bottleneck.
ii. Data Integration Complexity: Combining datasets from multiple fragmented government agencies and international organisations requires complex data-sharing mechanisms that are not yet fully standardised.
iii. Missing Variables: The current model excludes potentially vital data sources, such as high-resolution satellite imagery or social media sentiment analysis, which could further enhance early warning signals.

iv. Need for Continuous Updates: To remain relevant, the model requires constant re-training with new data to adapt to evolving environmental conditions and changes in human behaviour or sanitation infrastructure

3. **Machine Learning for Predicting Residual Malaria in Jazan, Saudi Arabia**

This case study which was based on research by Sadiq et al. (2024), explores the application of machine learning (ML) to predict residual malaria infections in the Jazan region of Saudi Arabia. This research is particularly significant as it focuses on a low malaria-transmission region, aiming to use environmental factors to bolster eradication efforts.

The researchers implemented a comparative framework to determine which model best captures the relationship between malaria incidence and climate with the following key implementations:

i. Data Integration: The study utilized a dataset of malaria infections confirmed via microscopic examination of blood smears (the gold standard) from 2015 to 2020. These records were combined with numerical meteorological data, including average temperature, maximum temperature, and relative humidity.

ii. Methodology: The dataset was split into an 80:20 ratio for training and testing/validation. The implementation involved three distinct ML models: Artificial Neural Networks (ANN), Random Forest Regression (RFR), and Regularized Linear Regression (RLR).

iii. Algorithm Performance: The findings suggested that the ANN model was the most effective for this task. This is attributed to the ANN's superior ability to capture complex, non-linear relationships inherent in climate-disease data, which traditional linear models or even some tree-based models might miss

Key Successes

i. Superior Accuracy: The ANN provided better predictions than other methods, evidenced by the lowest error across evaluation metrics such as Mean Squared Error (MSE).

ii. Early Warning Potential: The study successfully demonstrated that monitoring environmental variables in real-time can allow authorities to respond swiftly to emerging threats, potentially preventing epidemics before they occur.

iii. Optimized Resource Allocation: By identifying regional risk factors, the model provides public health authorities with a data-driven basis for directing resources (e.g., medical supplies and vector control) to where they are most needed.
iv. Capturing Complexity: The implementation highlighted the strength of AI in dealing with the complex biological and environmental interactions that define malaria transmission in a specific region.

Possible Misses and Limitations

Despite the technical successes, the study identified several "misses" and barriers to broader application:

i. Region-Specific Constraints: The findings are highly region-specific to Jazan; the model may not perform accurately if applied to areas with different environmental or ecological conditions without further localized training.
ii. Data Quality Issues: The model's accuracy was limited by the quality of data sourced from external databases, which suffered from inconsistencies or a lack of localized factors.
iii. Excluded Variables: A significant "miss" was the focus solely on meteorological factors. The researchers noted that the model excluded other critical variables, such as socio-economic conditions, human mobility, and specific vector control measures already in place.
iv. Technical Risks (Overfitting): While the ANN was flexible and accurate, the study warned that such models are prone to overfitting if not tuned with extreme care, which can compromise their reliability on new, unseen data.
v. Static Forecasts: The long-term forecasts generated by the system may fail to account for future shifts in healthcare infrastructure or changes in national medical interventions

4. **Malaria outbreak prediction in The Gambia**

This case study, based on the research by Khan et al. (2024), evaluates the implementation of machine learning (ML) techniques to predict malaria outbreaks across every district in The Gambia. The study is a pioneering effort in the region, aiming to provide health authorities with an early warning system to mitigate the high seasonal malaria prevalence, which typically peaks during and immediately after the rainy season.

The researchers developed a predictive framework by integrating two distinct longitudinal datasets spanning from 2013 to 2021. The model utilized historical meteorological data (rainfall, relative humidity, and maximum/minimum temperatures) and clinical data (monthly malaria cases and deaths) sourced from the Gambian Health Management Information System. In this research, a key aspect of the implementation was the inclusion of non-climatic features, specifically district-level population size and the month of the year, which were found to be significant in predicting when epidemics would occur.

The study compared the performance of eight supervised learning algorithms: C5.0 Decision Trees (DT), Artificial Neural Networks (ANN), K-Nearest Neighbors (KNN), Logistic Regression (LR), Extreme Gradient Boosting (XGBoost), Random Forest (RF), and Support Vector Machines (SVM) with both linear and radial kernels. To ensure reliability and prevent overfitting, the models were trained using repeated 10-fold cross-validation (repeated five times).

Key Successes

The implementation demonstrated that machine learning can provide highly accurate district-specific forecasts for malaria outbreaks as follows:

i. Top Performers: XGBoost and C5.0 Decision Trees emerged as the most effective models for the Gambian context.
ii. High Performance Metrics: On the testing dataset, both top models achieved a classification accuracy of 93.3% and a sensitivity (recall) of 96.7%. High sensitivity is particularly valuable in public health as it ensures that very few potential outbreaks are missed.
iii. Discriminatory Power: XGBoost demonstrated the highest capability to distinguish between outbreak and non-outbreak periods, with an Area Under the Curve (AUC) of 0.97.
iv. Pattern Recognition: The models successfully captured the complex, non-linear interplay between environmental variables, identifying that the highest malaria incidence consistently occurs between August and December.

Possible Misses and Limitations

Despite the technical successes, the study identified several "misses" and systemic challenges:

i. Data Imbalance: A major hurdle was the significant imbalance in the dataset, where "no outbreak" periods far outnumbered "outbreak" events, a common issue in epidemiological modeling that can bias results.
ii. Excluded Biological/Intervention Factors: The current study did not address other critical predictors such as the use of insecticide-treated bed nets, the coverage of indoor residual spraying, or the emergence of drug resistance.
iii. Generalization Challenges: The findings are region-specific; because climatic impacts vary significantly by geography, the models may require extensive localized retraining before they can be applied to different countries.
iv. Methodological Inconsistencies: The researchers noted that differences in datasets and reporting methods across the literature make it difficult to compare these results directly with other global malaria prediction studies

2.4 Integration and Architecture

This section outlines how the proposed machine learning-powered surveillance system for the prediction and visualization of malaria and cholera can be integrated with existing sources of data and implemented with the appropriate architectures of system implementation. The architectural frameworks of system architectures that are robust and support scalability, multimodal datasets, processing in real-time, and effective visualization necessary for public health response and decision-making are also explored. Architectural frameworks and integration strategies that are essential for connecting advanced machine learning predictive models with the existing public health systems and infrastructure are highlighted as well.

The integration and architecture of a machine learning-powered surveillance system for malaria and cholera in Kenya rely on multi-layered frameworks that bridge data collection, processing, and real-time user interaction (Kiama et al., 2023). Implementing these systems involves several technical options and architectural designs to ensure scalability, security, and interoperability with existing public health infrastructure.

2.4.1 System Integration Approaches

The proposed machine learning-powered surveillance system requires proper integration of multimodal data sources and services in order to achieve accurate prediction and visualization of the outbreaks of malaria and cholera in Kenya. Traditional surveillance systems for the prediction and visualization of infectious diseases pose challenges such as slow reporting times, limited predictive capability and no real-time integration of climate + health data (Maddah et al., 2023). The limitations posed by traditional infectious diseases surveillance systems such as Kenya Health Information System (KHIS) and DHIS2 underscore the need for more advanced predictive models that can be integrated with the existing public health infrastructure and multimodal datasets to improve response times in the management of malaria and cholera outbreaks in Kenya.

Integrating complex machine learning (ML) models into functional public health applications requires seamless data flow between various tools and platforms.

2.4.1.1 Integration with Existing Health Information Systems

The machine learning-powered surveillance system can be integrated with existing platforms such as the Kenya Health Information System (KHIS) to obtain historical disease records and the District Health Information System (DHIS)to provide predictive monitoring and tracking of outbreaks. This allows automatic or semi-automated data exchange between hospitals, county health departments, and the predictive system.

2.4.1.2 Integration with Weather and Climate APIs

Application Programming Interfaces (APIs) are essential for allowing different systems to share data and interoperate (Muriithi et al., 2024). Real-time environmental data such as rainfall, temperature, and humidity can be integrated using external APIs such as OpenWeatherMap. These data inputs are critical for predicting malaria and cholera outbreaks since both diseases are strongly influenced by climatic conditions.

2.4.1.3 Integration with GIS Platforms

Integrating GIS technology allows for the visualization of spatial distributions and the mapping of transmission "hotspots" to guide targeted resource allocation (Anthony, 2023). The system will integrate Geographic Information System (GIS) tools such as Leaflet.js or Google Maps API to visualize disease distribution across Kenyan counties. This enables spatial representation of outbreak hotspots for decision-making.

2.4.1.4 API-Driven Connectivity

A RESTful API architecture will be used to connect the frontend, backend, machine learning model, and external data sources. For instance, ML models developed in Python using the Flask framework can be connected to user-facing interfaces built with libraries like React. This approach allows developers to integrate AI technology regardless of their specific expertise in machine learning. This ensures seamless communication between system components and supports real-time data exchange.

2.4.2 System Architecture Options

There are different design architectures that can be adopted in the implementation of the proposed system. A robust and effective framework for disease outbreak prediction typically follows a structured, modular design to allow for flexibility and rapid development and often include specific core functions such as detection, reporting, confirmation, and feedback mechanisms. They must also address "cross-border" collaboration and multisectoral coordination to be effective in large-scale public health scenarios The most suitable system architecture options are microservices architecture and the layered architecture.

2.4.2.1 Layered Architecture (3-Tier Architecture)

The layered architecture is a common approach for full-stack systems such as the proposed machine learning-powered surveillance system. This architecture consists of three functional layers which work in coordination to deliver the system functionalities as follows:

i. Presentation Layer (Frontend): This layer consists of the frontend including the user interface for map visualization, analytics of predictions and dashboards.
ii. Application Layer (Backend): This is the system layer that handles the business logic of the system, processing of the data and communication with the predictive machine learning model.
iii. Data Layer (Database): This system architecture layer consists of the database which stores data such as the historical diseases data, environmental data and the results from the prediction.

The layered system architecture in the implementation of the proposed system will be suitable since it simple to design and implement, easy to maintain and debug errors as well as having a clear separation of concerns in the implementation of the system. However, this architecture has limitations such as complexity issues when integrating many services crucial for the system implementation.

2.4.2.2 Microservices Architecture

In this system architecture, the proposed system can be implemented into independent divisions which represents the independent system services such as the prediction service, data ingestion service and the visualization service. Each system service communicates with the use of APIs. The microservices system architecture is highly scalable and flexible even for large systems, easier to update individual system components and suitable for real-time and large-scale systems . On the flip side, the microservices system architecture poses complexity in the design and management, and requires strong API management and deployment strategies.

2.4.2.3 Hybrid Architecture

The proposed machine learning-powered surveillance system can adopt a hybrid architecture combining layered system architecture and the microservices approaches. This approach ensures both scalability and simplicity in the implementation of the system. An example of a hybrid system architecture can as follow this approach:

i. Core system (frontend, backend, database) uses layered architecture
ii. Machine learning and external integrations operate as microservices

2.5 Summary

The literature review reveals a significant shift in infectious disease surveillance, transitioning from traditional reactive systems to proactive, predictive data-driven frameworks powered by machine learning (ML) and artificial intelligence (AI).

The literature identifies a paradigm shift where traditional models are being supplemented or replaced by advanced computational techniques. The review demonstrate that Statistical and Epidemiological models hare historically the foundational models that excel at describing the biological processes but struggle with non-linear climate-disease interaction, also lacking real-time integration capabilities.

Studies suggest that supervised machine learning is the most efficient paradigm for climate-sensitive diseases in regions with limited resources. Supervised machine learning consistently outperforms traditional models through capturing complex patterns in multimodal datasets.

Deep Learning (Architectural Intelligence) utilizes architechtures like CNNs for spatial analysis and LSTMs for temporal sequences, thereby achieving diagnostic accuracies of up to 97.06% in malaria parasite detection.

Hybrid and Network Models are emerging schools of thought focusing on fusing mechanistic biological models with AI to maintain interpretability while enhancing predictive power.

Recent studies and implementations relevant to the application of machine learning in the prediction of infectious diseases demonstrate the practical success and limitations of these technologies across Africa as highlighted below:

1. In Kenya, a study about machine learning in the prediction of malaria and cholera by Muriithi et al. (2024) identified Random Forest machine learning algorithm as the most robust model for classifying test results, achieving 97.33% accuracy. Random Forest algorithm ranked the presence of Plasmodium falciparum and geographic region as critical predictors.
2. In Nigeria, research by Omankwu and Enefiok (2024) about machine learning in the prediction of cholera outbreaks developed a framework using Random Forest that achieved 92% accuracy. A major success was the integration of a user-friendly GIS interface for real-time alerts.
3. In Tanzania, the Adaptive Reference Machine Learning (ARML) model leveraged XGBoost to achieve 78.5% accuracy. It utilized ADASYN to handle class imbalance and PCA for dimensionality reduction.
4. In Gambia, the first study of its kind in the region used XGBoost and C5.0 Decision Trees to achieve 93.3% accuracy in predicting district-level outbreaks, emphasizing the importance of both climatic and non-climatic features like population size.

2.6 Research gaps

1. **Geographical Granularity and Lack of Actionable Visualization**

   In Kenya, regional health surveillance remains primarily descriptive, often lacking the granularity needed for county-level decision-making. Surveillance systems are frequently disconnected, leading to limited information at the sub-county level and delays in reporting.

   This research seeks to address this by developing a web-based predictive analytics system designed to visualize disease trends across all 47 Kenyan counties. By integrating Geographic Information Systems (GIS) and interactive dashboards, the system provides real-time geospatial visualizations of "hotspots" to aid health officials in efficient resource allocation.

2. **Absence of Simultaneous Multi-Disease Surveillance**
   There is a critical research gap in developing automated frameworks capable of providing early warning signals for both malaria and cholera simultaneously within a single platform.
   The general objective of this research is to develop a unified machine learning-powered surveillance system specifically tailored for predicting and visualizing outbreaks of both diseases across the Kenyan landscape.
3. **Technical Barriers (Data Imbalance and Sparsity)**
   Epidemiological modeling in developing countries is severely hindered by "broken" datasets characterized by missing information, data inconsistency, and extreme class imbalances—where negative cases can represent up to 96% of the data. Traditional models fail to perform adequately under these conditions, often leading to biased conclusions.
   The research proposal addresses this through a rigorous Data Preprocessing and Reengineering phase. It employs advanced sampling and balancing techniques—such as ADASYN or SMOTE—to restore sampling balance and handle dataset complexities before model training begins.
4. **Reactive Systems and Dependency on "Lagging Indicators"**
   Traditional surveillance systems in Kenya, such as DHIS2 and HMIS, are primarily reactive, meaning they rely on historical data and "lagging indicators" where cases are only recorded after patients present at clinics. By the time this data is aggregated and analyzed, outbreaks are often already in an exponential growth phase, leading to preventable loss of life and overstretched facilities.
   This research seeks to resolve this by implementing a proactive intervention strategy. It utilizes Long Short-Term Memory (LSTM) neural networks to identify patterns between real-time environmental anomalies (such as temperature and rainfall) and subsequent disease spikes, allowing medical resources to be deployed before an outbreak peak.

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APPENDICES

Appendix I: Project Resources

Hardware Requirements for the Project:

i. High-performance workstation/laptop.
ii. Cloud server (virtual machine).
iii. External storage devices (HDD/SSD).
iv. Uninterruptible power supply (UPS).
v. Networking equipment (router/modem).
vi. GPU-enabled machine (for machine learning training).

Software Resource Requirements for the Project:

i. Python – for data processing and machine learning development.
ii. Pandas – for data cleaning and manipulation.
iii. NumPy – for numerical computations.
iv. TensorFlow – for building and training ML models.
v. Keras – for designing neural networks (LSTM).
vi. Flask or FastAPI – for backend API development.
vii. React – for frontend user interface Leaflet or Mapbox – for GIS visualization.
viii. PostgreSQL (with PostGIS) or MySQL – for data storage.
ix. Git and GitHub – for version control and collaboration.

Appendix II: Project Budget

Table 1: Project Budget

Item	**Description**	**Quantity**	**Unit Cost (Ksh)**	**Total Cost (Ksh)**
Laptop	Development machine	1	60,000	60,000
Internet	Monthly subscription	3 months	2,000	6,000
Software	Licenses	-	0	0
Printing	Documentation	-	2,000	2,000
Miscellaneous	Contingency	-	2,000	2,000
Research costs	Research	3 months	2,000	6,000

Total Estimated Cost = Ksh. 76,000

Appendix III: Project Schedule

DURATION	Jan-26				Feb-26				Mar-26				Apr-26				May-26				Jun-26				Jul-26				Aug-26			
	Weeks				Weeks				Weeks				Weeks				Weeks				Weeks				Weeks				Weeks			
ACTIVITIES	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Draft Project																																
Project Proposal																																
Logical Design																																
Project Defense																																
Project Review																																
Detailed Design																																
Implementation																																
Testing																																
Project Documentation																																
Project Presentation																																

Figure 3: Project Gantt Chart